FurSight: A Mobile App for Geotagged Sighting and AI-Based Identification of Campus Cats and Dogs

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***Abstract*— Monitoring campus animals in university settings has become increasingly relevant as stray and resident cats and dogs become part of the community's daily environment. This study presents FurSight, a community-driven mobile application for the geotagged sighting and AI-based individual re-identification of campus cats and dogs at the University of the Philippines Los Baños (UPLB). FurSight enables users to submit geotagged photo sightings, which are processed on-device through a two-stage AI pipeline: a YOLOv8n model for animal detection and classification, and an EfficientNet-B0 model with cosine similarity matching for individual re-identification. Detected sightings are stored in a Firebase backend and visualized through a real-time heatmap. The detection model achieved a mean Average Precision at IoU 0.50 (mAP@0.5) of 0.9944, with precision of 0.9931 and recall of 0.9848. The re-identification model achieved an overall similarity accuracy of 0.9401, with same-animal pair accuracy of 0.9144 and different-animal pair accuracy of 0.9658. Usability was evaluated using the System Usability Scale (SUS) with 20 respondents from the UPLB community, yielding a mean SUS score of 83.88, which falls in the "Excellent" grade range. These results suggest that FurSight provides a usable and technically viable solution for community-based campus animal monitoring.**

***Index Terms*— Animal re-identification, campus animals, citizen science, Flutter, geotagged sighting, heatmap, mobile application, YOLOv8n, EfficientNet-B0**

I. Introduction

A. Background of the Study

Tracking local animals in urban and university settings has become increasingly important as interactions between humans and animals continue to rise. In academic institutions, such as universities, stray or resident animals often roam freely, becoming part of the community's daily environment. Understanding their movement patterns can provide insights into their behavior, typical locations, and potential impact on human activities. Traditionally, tracking animals has relied on GPS-based methods, but these approaches are costly and impractical for localized studies [1]. With the advancement of community-based tracking and artificial intelligence (AI)-assisted recognition, modern methods now enable efficient tracking through geotagged reports and image-based identification.

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Community participation in animal tracking has been widely explored in citizen science projects, where individuals contribute sightings and photographic evidence of animals to shared databases [2]. By leveraging smartphone cameras and geotagged observations, users can document the locations of animals, enabling researchers to analyze patterns over time. AI-driven image processing further enhances accuracy by identifying individual animals based on visual features such as coat patterns and body shape [3]. Recent studies emphasize the role of everyday individuals in contributing to scientific tracking efforts, demonstrating how crowdsourced data can lead to meaningful insights about animal movements [4].

In university settings, this approach can help determine common areas where animals are found, ensuring a non-intrusive yet effective way of mapping their presence. This study focuses on developing a community-driven tracking system that enables users to report animal sightings through photos and geotagged locations. The AI pipeline compares submitted images to determine whether they depict previously recorded individuals, and the system generates a real-time heatmap of sighting locations. By implementing this approach within the lower campus of the University of the Philippines Los Baños (UPLB), the study aims to provide an efficient and low-cost method for monitoring local animal movements.

B. Statement of the Problem

Currently, there is no centralized or organized method to monitor common animals within the UPLB campus. Sightings are anecdotal and undocumented, making it difficult to identify specific individuals, understand their movement patterns, or assess their well-being. This lack of data hinders efforts in animal welfare, public awareness, and even campus safety. There is a need for an accessible system that can record, identify, and track these animals found in the university.

C. Objectives of the Study

This study aims to develop a mobile application that allows users to report sightings of common animals, particularly cats and dogs, around the UPLB lower campus. The specific objectives are:

1) To establish a system that implements AI-based image recognition to detect and classify animals and identify

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individual animals based on visual similarity.
2) To record geotagged reports submitted by users and store them in a centralized database.
3) To visualize reported sighting data using a real-time heatmap to show areas with high animal activity.

D. Significance of the Study

This study is significant as it proposes a practical and community-powered solution to the problem of untracked animal populations in public spaces. By allowing students and members of the UPLB community to participate in data collection, the app encourages shared responsibility and awareness for animal welfare. Additionally, researchers, animal advocates, and campus administrators can benefit from having a data-driven understanding of animal movements, which can support initiatives in health, safety, and animal care.

E. Scope and Limitations

The study focuses solely on common animals in the UPLB lower campus, particularly cats and dogs. To ensure the quality and accuracy of the AI-based image analysis, users are advised to take photos only between 8:00 AM and 5:00 PM, when natural lighting conditions are ideal. Poor lighting outside of these hours may lead to unclear or unusable images. Additionally, for the best results, the animal should be clearly visible in the photo, with key features such as the face, fur pattern, and body shape unobstructed. The AI system's accuracy depends heavily on the clarity and quality of the submitted images.

It is also important to note that the system recognizes and matches animals based solely on visual features extracted from submitted photos, and does not utilize tagging, tracking collars, or other physical forms of identification. The app is limited to user participation and may not reflect the full movement patterns or real-time location of animals. Furthermore, its performance will depend on the volume and consistency of user submissions over time. The re-identification model was trained on publicly available datasets and has not yet been fine-tuned on campus-specific animals, which may affect accuracy for individuals not well-represented in the training data.

II. Review of Related Literature

Tracking the movement and whereabouts of local animals in urban and semi-urban environments has become increasingly relevant for understanding their behavior and interactions with human communities. Unlike traditional wildlife tracking methods that rely on GPS devices, this study focuses on community-sourced tracking through smartphone-captured images, geotagged reports, and AI-based analysis. By integrating detection and re-identification models, this approach provides a cost-effective and scalable solution for monitoring common animals such as cats and dogs within the lower campus of the University of the Philippines Los Baños (UPLB).

Community-Based Animal Tracking and Citizen Science

Community-based animal tracking and citizen science initiatives have grown in importance for environmental monitoring and wildlife conservation. These initiatives rely on public participation to collect data, particularly through digital platforms and mobile applications. Projects such as iNaturalist enable users to capture images of animals, geotag them, and upload them to create a shared database [5]. This community-driven approach facilitates the mapping of species distributions, helping to identify animal hotspots and common movement patterns in urban and semi-urban environments.

Citizen science projects have proven effective in gathering large amounts of data, as individuals contribute observations on animals in their local environment [6]. The integration of smartphone cameras and geotagging technologies makes it easier for people to participate in tracking initiatives, resulting in more accurate and real-time data. However, a challenge arises in ensuring the accuracy and consistency of user-submitted data, especially in distinguishing between individual animals or species.

Animal Detection Models

Automated animal tracking begins with robust detection models capable of identifying animals in varied environments. The YOLO (You Only Look Once) family of models has established itself as a leading approach for real-time object detection, offering a balance between speed and accuracy that makes it well-suited for mobile deployment [7]. YOLOv8, the most recent iteration, introduces improved backbone architectures and anchor-free detection heads, achieving state-of-the-art performance on standard detection benchmarks while remaining efficient enough for edge deployment [8]. In contrast to earlier approaches such as SSD MobileNetV2, which employed a two-branch architecture of a single-shot detector with a lightweight backbone [9], YOLOv8 provides more accurate bounding box predictions, particularly for small or partially visible objects — a common challenge in campus animal photography where animals may be partially obscured by vegetation or captured from a distance.

Animal Re-Identification Models

Re-identification (re-ID) refers to the task of recognizing the same individual across different images, without relying on physical tags. This is inherently harder than classification, as it requires the model to learn fine-grained visual features that distinguish individuals of the same species. Deep metric learning approaches train neural networks to produce embedding vectors in which images of the same individual cluster closely together in vector space, while images of different individuals are pushed apart [10].

EfficientNet-B0, introduced by Tan and Le [11], achieves strong feature extraction performance through compound scaling of depth, width, and resolution. When used as the backbone of a re-identification model trained with a combination of cross-entropy loss and triplet loss, EfficientNet-B0 learns discriminative embeddings that are suitable for cosine similarity

matching at inference time. Triplet loss training, in particular, has been shown to improve re-ID accuracy in animal identification tasks by explicitly optimizing the relative distances between same-individual and different-individual image pairs [12]. This approach is adopted in FurSight to match submitted animal photos against stored embeddings in Firebase.

Geotagging and Real-Time Mapping

Integrating geotagged images with AI models allows for near real-time mapping of animals' movements. Geographic Information System (GIS) tools such as ArcGIS and open-source mapping platforms like OpenStreetMap (OSM) have been used in similar projects for visualizing spatial data [13]. Community-driven applications such as iNaturalist use geotagged observations to document species distributions [5].

Similarity Threshold in Image Comparison

In animal re-identification, determining the similarity between image embeddings is crucial to accurately match individuals across different photos. The similarity threshold defines whether two images depict the same animal, based on the cosine similarity of their embedding vectors. Studies on deep metric learning for animal re-identification highlight that thresholds must account for factors such as lighting, angle, and image resolution, as poor-quality images reduce matching accuracy [14] [15].

While thresholds are often calibrated to balance precision and recall, research on photo-identification systems emphasizes that selecting an optimal threshold involves trade-offs between true positives and false positives [16]. Higher similarity thresholds reduce false positives but risk missing valid matches, whereas lower thresholds increase recall at the expense of precision. In FurSight, a similarity threshold of 0.55 was empirically determined during on-campus physical device testing, chosen to balance the rate of incorrect animal merges against the rate of missed matches. An additional embedding confidence gate of 0.70 is applied when updating an individual's average embedding, providing a second layer of protection against noisy updates from low-quality submissions.

Ethical Considerations in Animal Tracking

Although this study does not involve direct handling or tagging of animals, ethical considerations must be addressed. The Philippine Animal Welfare Act (RA 8485) outlines the legal framework for conducting research involving animals [17]. Additionally, ensuring that data collected through community reports remains ethical and does not disrupt the animals' natural behavior is a crucial aspect of responsible tracking [18].

Challenges and Limitations of Community-Sourced Tracking

While community-based tracking offers a cost-effective and scalable solution, it also presents several challenges. One major limitation is the accuracy of data collected from untrained contributors, as the quality of image submissions can vary greatly. For instance, in the Wildwatch Kenya project, community scientists were 83.4% accurate overall, but often missed animals that were distant or difficult to see, leading to many false empty images rather than misidentifications of species present [19]. Similarly, a University of Michigan study found that crowdsourced wildlife identification can be highly accurate (up to 97%) when properly designed, but some species are harder to distinguish, highlighting variability in contributor skill and data quality [20].

Factors such as poor lighting or blurry photos can hinder the AI model's ability to recognize features accurately. Research shows that low-light or motion-blurred images reduce both human and AI recognition accuracy, emphasizing the need for high-quality, well-lit images for reliable identification [21] [19]. To mitigate this, it is essential to set guidelines for photo quality, such as limiting image submissions to before 5 pm to avoid low-light conditions. Another limitation is the distance from which the animal is photographed. For optimal recognition, animals should be captured at a reasonable distance where their features are clearly visible. Studies from community science projects demonstrate that animals photographed too far away are often missed or misclassified by volunteers and AI alike, as key features become indistinguishable at greater distances [19]. If the animal is too far away, the AI may struggle to differentiate it from other animals or objects in the image.

These limitations should be considered when designing the user experience and AI model to improve the reliability and accuracy of the system, including implementing submission guidelines and possibly showing image bursts rather than single shots to help observers detect subtle movement and presence [19].

Practical Applications and Impact on the Local Community

Tracking local animal movements in a university campus setting has multiple benefits. Understanding where animals are commonly found can help improve campus planning, ensure safer interactions between humans and animals, and support environmental awareness efforts. For example, a study of wildlife habitat management across 60 college campuses in the U.S. found that many institutions successfully implement wildlife habitat projects that enhance biodiversity and sustainability with relatively low costs, demonstrating the feasibility and benefits of such initiatives in higher education settings [22]. Similarly, student-led projects using camera traps on campuses have enabled hands-on learning in field biology while generating valuable data on species richness, abundance, and habitat use, thereby fostering environmental stewardship among the university community [23].

The integration of AI models for animal detection, classification, and re-identification provides a robust framework for community-driven tracking. Leveraging smartphone-based image collection, geotagging, and deep learning techniques enables efficient and ethical monitoring of local animals in a university campus setting, as demonstrated by ongoing projects that combine remote camera technology with student participation to map wildlife presence and behavior [24] [25].

Future studies should explore optimizing re-identification models to handle variations in image quality and distance,

as well as sustaining long-term engagement of community participants in animal tracking initiatives, which is essential for maximizing the impact of such programs on campus biodiversity conservation and education [26].

III. Materials and Methods

A. Development Tools

The mobile application was developed using Flutter for the front end and Firebase for the back end and database. The Geolocator package retrieves the user's GPS location when they upload or take a photo. The application utilizes the Google Maps API to generate a heatmap visualizing reported sighting locations within the UPLB lower campus, rendered within the mobile application through a WebView component. All model training was conducted on Google Colab using T4 GPU instances. The system was developed on a computer with the following specifications:

- 8th Gen Intel(R) Core(TM) i5-8265U CPU @ 1.60GHz 1.80 GHz
- 16.00 GB RAM
- 256 GB SSD
- 64-bit operating system, x64-based processor
- Windows 11

B. AI Models and Dataset

The application employs a two-stage AI pipeline to process user-submitted images: a detection stage that locates and classifies the animal in the image, and a re-identification stage that determines whether the detected animal matches a previously recorded individual.

1) Animal Detection and Classification — YOLOv8n: For object detection and animal classification, this study uses YOLOv8n (You Only Look Once, version 8 nano), the smallest and fastest variant of the YOLOv8 architecture developed by Ultralytics [8]. YOLOv8n is optimized for real-time inference on mobile and embedded devices while maintaining strong detection accuracy. It detects the presence of cats or dogs in the submitted image and outputs bounding box coordinates and confidence scores.

The detection model was trained on the Dog and Cat Detection dataset sourced from Kaggle (andrewmvd/dog-and-cat-detection) [27], which contains 3,686 images with Pascal VOC XML annotations. Annotations were converted to YOLO format, and the dataset was split into training (80%), validation (10%), and test (10%) sets with a fixed random seed for reproducibility. The model was trained for 50 epochs with image size 416×416, batch size 32, early stopping patience of 10, and data augmentation enabled. Training was performed on Google Colab using a T4 GPU.

After detection, the bounding box coordinates are used to crop the animal region from the original image. This cropped region is then passed to the re-identification stage.

2) Animal Re-Identification — EfficientNet-B0 with Triplet Loss: To determine whether a detected animal has been previously recorded, the application uses an EfficientNet-B0 backbone with a custom embedding head trained using a combined cross-entropy and triplet loss objective. EfficientNet-B0 was chosen for its strong feature extraction capability relative to its model size, producing a 128-dimensional embedding vector for each input image crop [11]. At inference time, this embedding is compared against stored embeddings using cosine similarity. If the best-matching score meets or exceeds the similarity threshold, the submitted animal is matched to the existing individual; otherwise, it is registered as a new individual.

The re-identification model was trained on two subsets from the WildlifeReID-10k dataset [28], sourced from Kaggle (wildlifedatasets/wildlifereid-10k):

- CatIndividualImages — 509 individual cats, used for the cat re-ID component.
- DogFaceNet — 1,001 individual dogs, used for the dog re-ID component.

Only individuals with at least 5 images were retained, yielding a total of 1,510 individuals and 20,061 images. The dataset was split 80/20 into training and validation sets per individual. Model training was performed for 60 epochs with a 5-epoch warmup schedule, using a PKSampler to ensure that each batch contained a fixed number of identities with multiple images each — a requirement for effective triplet mining. Training used a combined cross-entropy and triplet loss, with the Adam optimizer starting at a learning rate of 1×10^{-3} and decaying via cosine annealing.

These models are selected for their efficiency and compatibility with mobile platforms, ensuring that the application can perform complex image analysis tasks in real-time while maintaining responsiveness and conserving device resources.

C. System Architecture and Pipeline

The AI inference pipeline, as can be seen in Figure 1, executes entirely on the user's device in the following sequence:

1) The user captures or selects a photo from their gallery.
2) The image is resized to 416×416 pixels and fed into YOLOv8n. Detections with confidence below 0.15 are discarded.
3) The highest-confidence detection's bounding box is used to crop the animal region from the original image.
4) The cropped region is resized to 224×224 pixels, normalized using ImageNet mean and standard deviation values, and fed into EfficientNet-B0 to produce a 128-dimensional embedding vector.
5) The embedding is compared via cosine similarity against all stored embeddings retrieved from Firebase Firestore. If the best match scores ≥ 0.55, the animal is identified as a known individual; otherwise, it is treated as a new entry.
6) If the matched similarity is ≥ 0.70, the individual's stored average embedding in Firestore is updated using

a running average, with a Firestore transaction ensuring race condition safety.

7) Only after on-device inference is complete does the app communicate with the internet: the photo is uploaded to Firebase Storage, and the sighting record — including the animal's identity, GPS coordinates, and timestamp — is written to Firestore. The heatmap updates on next load.

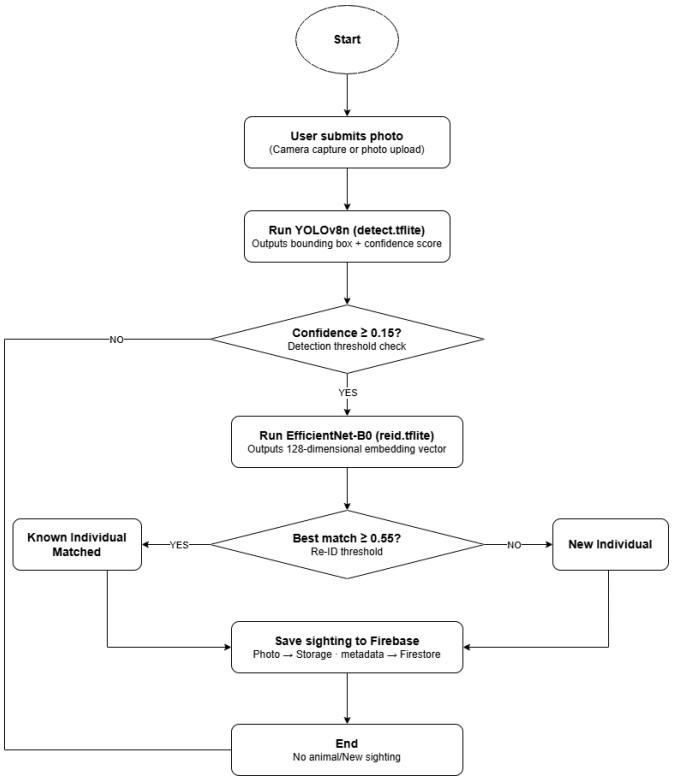


Fig. 1: Flowchart for animal detection, classification, and re-identification

D. Application Features

FurSight is a community-driven mobile application with the following core features:

1) User Authentication: Firebase Authentication handles user registration and login, ensuring that sighting data is tied to individual contributor accounts.

2) Sighting Submission: Users can submit animal sightings by taking a photo with the device camera or uploading from the gallery. The app automatically retrieves GPS coordinates from the device. The submission triggers the on-device AI pipeline described above.

3) Community Feed: A scrollable feed of recent sightings submitted by all users, displaying the identified animal, detection confidence, sighting location, and timestamp.

4) Heatmap: A Google Maps-based heatmap rendered via WebView, showing sighting density across the UPLB lower campus. Users can filter the heatmap by animal type.

5) Submission Logs: Users can review their own past submissions, including the AI identification result and location data.

6) Animal Profiles: Each identified individual in the database has a profile showing its name, type, and sighting history.

E. User Evaluation

To evaluate the overall usability of the mobile application, the study used the System Usability Scale (SUS), a validated 10-item questionnaire that yields a single composite score from 0 to 100 [29] [30]. The SUS provides a reliable measure of perceived ease of use, learnability, and overall satisfaction and has been widely applied in mobile application usability research. A minimum of 20 respondents from the UPLB community were required to use the app and complete the SUS survey, which was administered online through Google Forms.

In addition to the SUS items, the survey collected data on task difficulty ratings across 11 app screens, feature ease-of-use ratings across 6 features, AI detection and re-identification accuracy ratings, input method comparison, retake frequency, perceived factors affecting AI accuracy, and open-ended feedback on useful features, confusing elements, and suggested improvements. This supplementary data provides qualitative and quantitative context beyond the SUS composite score, helping to identify specific areas for improvement in the app's design and AI pipeline.

IV. Results and Discussion

A. Developed Mobile Application

FurSight is a community-driven Android application built with Flutter, backed by Firebase Authentication, Cloud Firestore, and Firebase Storage. Upon launching the app, users are directed to the landing screen, which offers options to log in or create an account. The login screen accepts registered credentials; invalid input is flagged immediately via a snackbar notification. New users complete a brief registration form requiring a name, email, and password, and must acknowledge the terms and privacy policy before their account is created.

After logging in, users are guided through a short onboarding flow that explains the app's purpose and submission guidelines. The main dashboard is the first screen displayed on subsequent launches. It shows a summary of recent community sightings, provides quick access to all core features, and displays a prompt to begin submitting a sighting.

The core sighting submission flow allows users to either capture a photo using the device camera or upload one from their gallery. Once a photo is selected, the on-device AI pipeline executes automatically: YOLOv8n detects and classifies the animal, EfficientNet-B0 generates an embedding, and cosine similarity matching against stored embeddings determines whether the animal is a known individual or a new entry. The result — including the detected animal type, matched individual name or new-individual status, and similarity confidence — is displayed to the user before submission is confirmed. GPS coordinates are retrieved from the device at

the time of capture and attached to the sighting record, which is then written to Firestore and its photo uploaded to Firebase Storage.

The community sightings feed displays geotagged sighting cards from all users in reverse chronological order. Each card shows the submitted photo, the identified animal, sighting location, and timestamp. The heatmap feature renders a Google Maps-based density map via a WebView component, visualizing where sightings are concentrated across the UPLB lower campus. Users can switch between a general heatmap view and an individual-animal focus mode that shows pins for a specific animal's sighting history. The submission logs screen allows users to review their own past contributions. Each identified campus animal has a profile page showing its name, species type, and a chronological list of sightings. The How It Works screen provides a plain-language explanation of the submission process and AI pipeline, and the Emergency Hotlines screen lists relevant campus and local contacts for animal welfare situations.

Screenshots of the developed application are provided in the Appendix.

B. AI Model Performance

1) Detection Model — YOLOv8n: The YOLOv8n detection model was trained for 50 epochs on the Dog and Cat Detection dataset [27], consisting of 3,686 annotated images split 80/10/10 into training, validation, and test sets. Training was conducted on Google Colab using a T4 GPU with an image size of 416×416 pixels, batch size of 32, and early stopping patience of 10. Table I presents the evaluation results on the test set.

TABLE I: YOLOv8n Detection Model Performance on Test Set

Metric	Value
mAP@0.5	0.9944
mAP@0.5:0.95	0.9115
Precision	0.9931
Recall	0.9848

The model achieved a mean Average Precision at IoU threshold 0.50 (mAP@0.5) of 0.9944, with precision of 0.9931 and recall of 0.9848. These results substantially exceed the target threshold of mAP@0.5 $\geq$ 0.80 set during development and demonstrate strong generalization on unseen images. The high recall value indicates that the model misses very few actual animals in submitted photos, which is the more important property for this application: a missed detection results in no sighting being recorded at all, whereas a low-confidence false positive is filtered out by the 0.15 confidence threshold before any downstream processing occurs.

The strong detection performance is consistent with prior work demonstrating YOLOv8n's effectiveness for mobile-optimized detection tasks [8]. After export to TFLite with int8 quantization, the model occupies 3.08 MB on disk, making it practical for on-device inference without requiring network access.

2) Re-Identification Model — EfficientNet-B0: The EfficientNet-B0 re-identification model was trained for 60 epochs on subsets of the WildlifeReID-10k dataset [28], comprising 509 individual cats (CatIndividualImages) and 1,001 individual dogs (DogFaceNet), for a combined training set of 1,510 individuals and 20,061 images. Training used a combined cross-entropy and BatchHard triplet loss [12] with a PKSampler (P identities, K images per identity per batch), an Adam optimizer starting at a learning rate of 1×10^{-3} with cosine annealing decay, and a 5-epoch linear warmup. Table II presents the similarity evaluation results on the validation set.

TABLE II: EfficientNet-B0 Re-Identification Model Performance on Validation Set

Metric	Value
Same-animal pair accuracy	0.9144
Different-animal pair accuracy	0.9658
Overall similarity accuracy	0.9401
Similarity threshold used	0.40

The model achieved an overall similarity accuracy of 0.9401 at a threshold of 0.40, meaning that 94.01% of image pairs were correctly identified as either the same individual or different individuals. Same-animal pair accuracy of 0.9144 indicates that the model correctly recognized the same campus animal across different photos in 91.44% of cases. Different-animal pair accuracy of 0.9658 indicates that 96.58% of image pairs showing two different individuals were correctly distinguished. These results are consistent with findings in the animal re-identification literature, which demonstrate that deep metric learning with triplet loss produces embeddings capable of reliably separating individuals within the same species [14].

The model exports to a 9.22 MB TFLite file. At inference time, each image crop is resized to 224×224 pixels, normalized using ImageNet mean and standard deviation values, and processed within a few hundred milliseconds on a mid-range Android device. The on-device threshold used in the deployed app (0.55) is higher than the validation threshold (0.40), as it was empirically tuned during on-campus physical device testing to reduce false merges between visually similar but distinct campus animals.

C. Usability Evaluation

1) Respondent Profile: A total of 20 respondents from the UPLB community participated in the usability evaluation of FurSight. All respondents were students affiliated with the university. The majority (18 out of 20, or 90%) had no prior experience with a similar animal tracking or sighting application, which provides a more conservative and externally valid assessment of the app's learnability and ease of use.

2) System Usability Scale Results: After interacting with the application and completing all assigned tasks, respondents answered the 10-item System Usability Scale (SUS) questionnaire [29]. SUS scores range from 0 to 100, where higher scores indicate greater perceived usability. The aggregate results are summarized in Table III.

TABLE III: SUS Scores

Mean (all 20)	**83.88**
Mean (excl. outlier)	**87.63**
Median	**85.00**

The mean SUS score across all 20 respondents was 83.88, with a median of 85.00. According to the adjective rating scale established by Bangor, Kortum, and Miller [30], a mean SUS score between 80.3 and 88.9 corresponds to a grade of B and is categorized as "Excellent." This places FurSight well above the commonly accepted threshold of 68, which represents average usability, and indicates that the majority of respondents found the application genuinely easy to use and well-designed.

One respondent obtained a notably low score of 12.5. Upon inspection of this respondent's raw item responses, a consistently inverted pattern was observed: all positively-worded SUS items received a rating of 1 and all negatively-worded items received a rating of 5, which is the exact inverse of the expected response direction for a satisfied user. This pattern, combined with the same respondent giving moderate likelihood-to-use and likelihood-to-recommend scores of 5 out of 10, strongly suggests a scale misunderstanding rather than genuine dissatisfaction. When this outlier is excluded, the mean SUS score rises to 87.63, which approaches the upper boundary of the "Excellent" range. Both figures are reported transparently; the full-sample mean of 83.88 is used as the primary reported result.

3) Task Difficulty Ratings: Respondents rated the difficulty of 11 tasks on a scale from 1 (very easy) to 5 (very hard). All tasks were rated below 2.0, indicating that no workflow presented meaningful difficulty to users. Table IV presents the mean rating per task.

TABLE IV: Mean Task Difficulty Ratings (1 = Very Easy, 5 = Very Hard)

Task	Mean
Viewing community sightings feed	1.10
Reviewing submission logs	1.10
Uploading a sighting from gallery	1.15
Navigating the How It Works screen	1.15
Finding Emergency Hotlines	1.15
Launching and understanding the Home screen	1.30
Submitting a sighting via camera	1.40
Exploring the Heatmap	1.20
Waiting for and understanding AI result	1.50
Interpreting detected animal type/re-ID	1.55
Understanding individual match and score	1.60

The highest-difficulty tasks were those related to interpreting the AI output — specifically understanding the individual animal match and the associated similarity score (mean 1.60) and interpreting the detected animal type and re-identification result (mean 1.55). While these ratings still fall comfortably in the "very easy" range, they point to the AI result display as the relative weak point in the interface, which is consistent with qualitative feedback discussed below.

4) Feature Ease-of-Use Ratings: Respondents also rated the ease of use of six application features on a scale from 1 (very hard) to 5 (very easy). All features scored above 4.0, with Submission Logs, the How It Works screen, and Emergency Hotlines receiving the highest ratings. Table V presents these results.

TABLE V: Mean Feature Ease-of-Use Ratings (1 = Very Hard, 5 = Very Easy)

Feature	Mean
Viewing Submission Logs	4.65
Navigating How It Works screen	4.65
Finding Emergency Hotlines	4.60
Viewing Community Sightings Feed	4.50
Submitting a Photo Sighting	4.45
Using the Heatmap	4.40

The Heatmap received the lowest ease-of-use rating (4.40), though this remains high in absolute terms. Qualitative feedback attributed this partly to the map not supporting rotation, which some users found disorienting when trying to orient themselves relative to the campus.

5) AI Accuracy Ratings and Qualitative Feedback: Respondents rated the accuracy of the detection and re-identification components separately on a scale from 1 to 5. Detection accuracy received a mean rating of 3.85 out of 5, while re-identification accuracy received a mean of 3.50 out of 5. Both scores are above the scale midpoint, indicating that users perceived the AI performance as acceptable, though with room for improvement. The lower rating for re-identification is expected, as individual identity matching is a significantly harder task than species-level detection.

Ten out of 20 respondents (50%) reported encountering at least one incorrect classification or identification, four were unsure, and six reported no errors. The most commonly reported error type was a cat being identified as a dog, with multiple respondents attributing this to challenging conditions such as dark or low-contrast backgrounds, sleeping or flat-lying postures that obscured facial features, and images where the animal occupied a small portion of the frame. One respondent noted that the system flagged a non-campus animal as a match to an existing individual, and another observed instances of a human being flagged as an animal, which are consistent with the low detection threshold (0.15) set to maximize recall.

When asked what factors most affected AI detection accuracy, respondents most frequently cited animal posture or angle (15 responses) and lighting conditions (15 responses), followed by camera focus or blur (13), distance from the animal (12), and image background noise (6). These factors align with the known limitations of image-based re-identification under real-world conditions [14] [15].

Regarding image input, the majority of respondents found camera capture and gallery upload to perform similarly (8 respondents), with 8 selecting "unsure" and only 4 expressing a preference for one input type over the other. Retake frequency was low overall: 11 respondents reported needing to retake or re-upload images only rarely, 3 reported never needing to, and 6 reported needing to do so sometimes or often.

6) Overall Satisfaction and Qualitative Feedback: Nine respondents rated FurSight as "Excellent" and 11 as "Good," with no respondents selecting a rating below Good. The mean likelihood-to-use score was 9.05 out of 10, and the mean likelihood-to-recommend score was 9.50 out of 10, indicating

strong user acceptance and willingness to adopt the application if it were officially released.

Qualitative feedback was predominantly positive. The heatmap and community sightings feed were most frequently cited as the most useful features, with respondents highlighting their practical value for locating specific campus animals and monitoring sighting patterns. Several respondents noted the community aspect of the app as particularly compelling. As one respondent expressed, the community sightings feed enables learning about UPLB's campus animals, while another described the collaborative nature of the platform as feeling like working alongside fellow animal lovers.

The most frequently cited usability issue was the ambiguity of the AI output display. Multiple respondents found the confidence percentage and similarity score labels unclear — specifically, several users were uncertain whether the confidence score referred to the certainty of the species detection or of the individual identity match. One respondent noted that the label "spotted with 90% confidence" was confusing because it was not clear whether the model was 90% confident about the species or about the individual being new to the database. This feedback highlights a specific and actionable UX improvement for future versions. Additional requested improvements included map rotation support, a pin-drop feature to manually specify a sighting location, clickable phone numbers on the Emergency Hotlines screen, a missing animal reporting feature, and a separate sign-in and sign-up screen.

V. Conclusion

This study presented FurSight, a community-driven mobile application for the geotagged sighting and AI-based individual re-identification of campus cats and dogs at the University of the Philippines Los Baños (UPLB). The application was developed using Flutter and Firebase, and implements a fully on-device two-stage AI pipeline consisting of a YOLOv8n detection model and an EfficientNet-B0 re-identification model.

The study successfully fulfilled all three stated objectives. First, an AI-based image recognition system was established that classifies submitted photos into cat or dog categories and identifies individual animals using deep metric learning with cosine similarity matching. The YOLOv8n detection model achieved a mAP@0.5 of 0.9944 with precision of 0.9931 and recall of 0.9848, while the EfficientNet-B0 re-identification model achieved an overall similarity accuracy of 0.9401 on the validation set. Second, geotagged sighting records submitted by users are stored in a centralized Firebase Firestore database, with photos archived in Firebase Storage. Third, sighting data is visualized through a real-time Google Maps heatmap rendered within the application, enabling the community to observe patterns of animal activity across the lower campus.

Usability evaluation using the System Usability Scale with 20 UPLB community respondents yielded a mean SUS score of 83.88, corresponding to an "Excellent" grade on the adjective rating scale established by Bangor et al. [30]. All task difficulty ratings were below 2.0 on a 5-point scale, and all feature ease-of-use ratings exceeded 4.0, confirming that the interface is accessible and intuitive for first-time users. Respondents expressed strong intent to use and recommend the application, with mean likelihood scores of 9.05 and 9.50 out of 10, respectively.

The primary limitation identified through testing is the clarity of the AI output display. Multiple respondents found the confidence and similarity score labels ambiguous, suggesting that the interface presentation of AI results requires refinement to be interpretable by non-technical users. Additional limitations include the dependence on image quality — particularly lighting conditions and animal posture — for accurate re-identification, and the use of a publicly available training dataset that does not include campus-specific animals, which may reduce re-identification accuracy for individuals not represented in the training distribution.

VI. Future Work

Based on the findings of this study, several directions are recommended for future development of FurSight.

The most immediately actionable improvement is the redesign of the AI result display. The confidence score and similarity percentage labels should be accompanied by plain-language explanations that distinguish between species-level detection confidence and individual identity matching. A brief tooltip or inline help text explaining what each value means would address the single most common usability complaint raised by respondents.

On the technical side, the re-identification model would benefit from fine-tuning on a dataset of campus-specific animals. The current model was trained on publicly available wildlife re-identification datasets and has not been exposed to the specific appearance, lighting conditions, and camera perspectives characteristic of the UPLB environment. Collecting a labeled dataset of campus cat and dog individuals over time — using confirmed sighting submissions as training data — would allow the model to be incrementally adapted to local conditions, improving matching accuracy for the animals most relevant to the application's users.

Additional feature improvements suggested by respondents that warrant consideration in future iterations include map rotation support to improve spatial orientation when browsing the heatmap, a manual pin-drop location input to allow users to specify a sighting location when GPS is unavailable or inaccurate, a missing animal reporting feature that would allow users to flag when a known individual has not been sighted in an extended period, clickable phone numbers on the Emergency Hotlines screen for direct dialing, and a separate sign-in and sign-up flow to reduce confusion for first-time users navigating the authentication screen.

From a research perspective, a longitudinal evaluation involving repeated use of the application over an extended period would provide more reliable data on re-identification accuracy under real-world conditions, long-term user engagement, and the utility of the heatmap for tracking animal movement patterns over time. Expanding the respondent pool and demographics — including faculty, staff, and non-CS students — would also strengthen the generalizability of the usability findings beyond the current sample.

APPENDIX

Application Screens

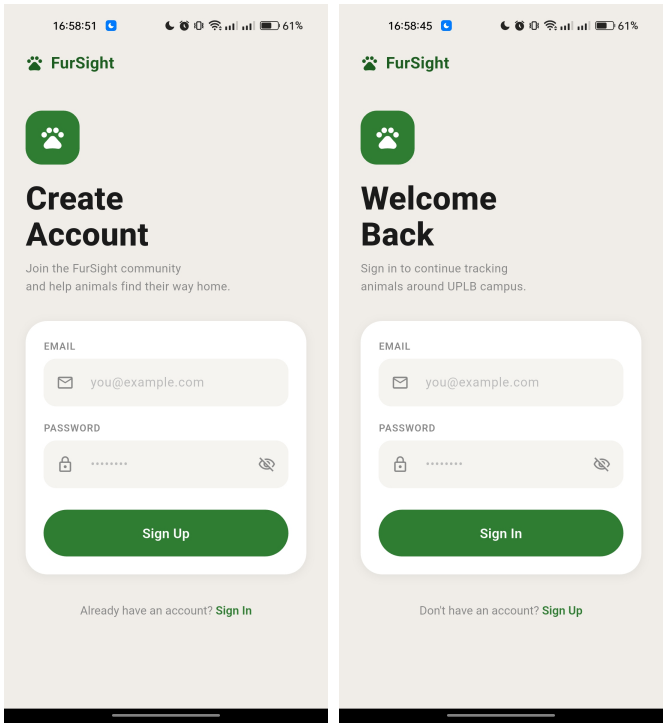


Fig. A1: Login and Sign-Up Screens

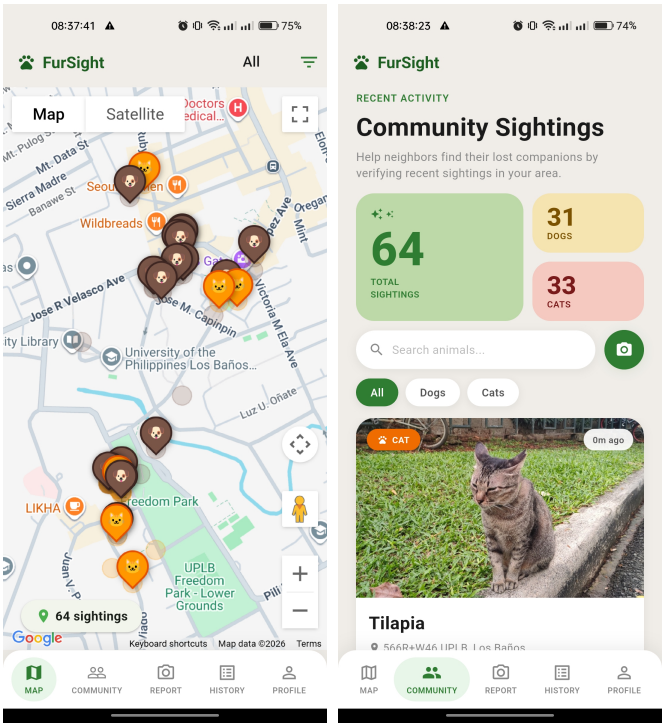


Fig. A2: Heatmap and Community Sightings Feed

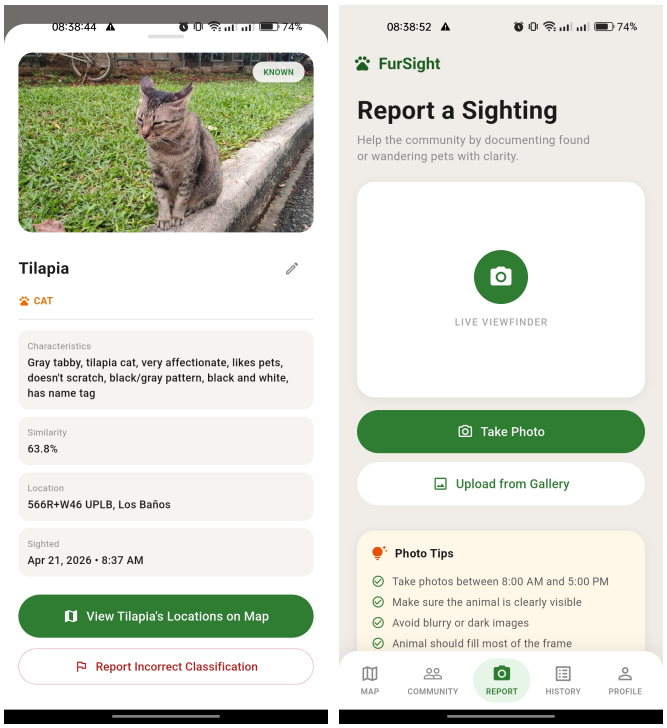


Fig. A3: Animal Profile Modal and Sighting Report Page

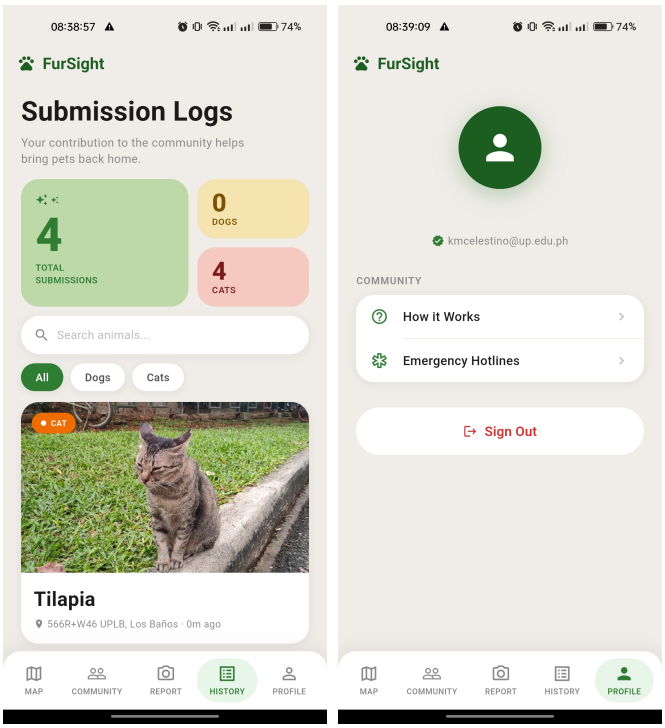


Fig. A4: Submission History and User Profile

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